

Continuous Random Variables

6.1 Probability density functions

Definition of Probability Density Function (PDF)

A.K.A density function, a continuous version of the PMF, and this function should always output a **non-negative real value**

The probability is $P(X \in A) = \int_A f(x)dx$

(a) $F(t) = \int_{-\infty}^t f(x)dx$

(b) $\int_{-\infty}^{\infty} f(x)dx = 1$

(c) If f is continuous, then $F'(x) = f(x)$, 即使 f 不連續, 在 f 連續的部分 $F'(x) = f(x)$

(d) For real number $a \leq b$, $P(a \leq X \leq b) = \int_a^b f(t)dt$

(e) $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b) = \int_a^b f(t)dt$

表示在連續分布中 $P(x) = 0$



Probability density functions

Ex 6.1 Experience has shown that while walking in a certain park, the time X , in minutes, between seeing 2 people smoking has a density function of the form

$$f(x) = \begin{cases} \lambda x e^{-x} & x > 0 \\ 0 & \text{otherwise.} \end{cases}$$

(a) Calculate the value of λ .
(b) Find the cumulative distribution function of X .
(c) What is the probability that Jeff, who has just seen a person smoking, will see another person smoking in 2 to 5 minutes? In at least 7 minutes?

p6.



Probability density functions

Sol:

$$(a) 1 = \int_{-\infty}^{\infty} f(x)dx = \int_0^{\infty} \lambda x e^{-x} dx = \lambda \int_0^{\infty} x e^{-x} dx.$$

By integration by parts, $\int x e^{-x} dx = -(x+1)e^{-x}$.

Thus $\lambda [-(x+1)e^{-x}]_0^{\infty} = 1$.

By l'Hopital's rule we get $\lim_{x \rightarrow \infty} (x+1)e^{-x} = \lim_{x \rightarrow \infty} \frac{x+1}{e^x} = \lim_{x \rightarrow \infty} \frac{1}{e^x} = 0$.

Therefore $\lambda [0 - (-1)] = 1$ or $\lambda = 1$.

p7.

